Classical Field Theory — Exercise Sheet 19

Adjoint identities, matter and surface currents

Hints and answers

All fields and coefficients on this sheet are smooth. Variations and gauge parameters in action identities have compact support inside the coordinate region. Use the Euler sign $\delta S = \int E_a \delta\phi^a$ and the connection conventions of Lecture 19. No equations of motion are imposed unless explicitly stated.

Exercise 19.1 (A second-order adjoint and the elementary converse). On an open coordinate patch $U \subset \mathbb{R}^{1+1}$ with coordinates (t, x), let a scalar parameter $\varepsilon \in C_c^\infty(U)$ produce two variations

$$R(\varepsilon)^1 = (1+tx)\varepsilon_{tt} + 2(t+x^2)\varepsilon_{tx} + e^t\varepsilon_x, \qquad R(\varepsilon)^2 = (1+t^2)\varepsilon_{xx} + x\varepsilon.$$

Compute $R^*(E_1, E_2)$, first in divergence form and then with every coefficient derivative expanded. Prove the compact pairing identity. Suppose an action has the compact first variation $\int E_a[\phi]h^a$ and $R^*_\phi E[\phi] = 0$ on U for every smooth field. Prove that $R_\phi(\varepsilon)$ is an infinitesimal gauge symmetry. Explain why the same converse holds for any finite-order operator $R_\phi(\varepsilon)^a = \sum_I r^{aI}_\alpha[\phi]\partial_I\varepsilon^\alpha$, without interpreting it as a converse for arbitrary conserved currents.

Hint. Integrate the mixed derivative twice, keeping both coefficient derivatives. The converse uses the same pairing in reverse; it requires an identity at every field, not only at solutions.

Answer.

$$R^*E = (1+tx)E_{1,tt} + 2(t+x^2)E_{1,tx} + 6xE_{1,t} + (2-e^t)E_{1,x} + (1+t^2)E_{2,xx} + xE_2.$$

The identity $R^*E = 0$ is sufficient for compact infinitesimal invariance.

Exercise 19.2 (The Euler term carried by charged matter). Write $A_{\mathrm{em}} = A_\mu \mathrm{d}x^\mu$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and let $u = (u_1, u_2)^{\mathsf{T}}$ be two real scalar fields. Given $q, \lambda, v > 0$, set

$$T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad D_\mu u = \partial_\mu u - qA_\mu Tu, \qquad L = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}D_\mu u \cdot D^\mu u - \frac{\lambda}{8}(|u|^2 - v^2)^2.$$

The gauge variations are $\delta A_\mu = \partial_\mu\varepsilon$ and $\delta u = q\varepsilon Tu$.

(a) Verify covariance of $D_\mu u$, derive both Euler arrays by compact variation, and derive the off-shell Noether identity containing them.

(b) On a coordinate patch take $A_\mu = 0$ and $u = (t, t^2)^{\mathsf{T}}$. Compute the ordinary divergence of the total gauge Euler array and its balancing scalar Euler term. Does the pure Maxwell identity remain an identity for that total array by itself?

(c) For a real orthogonal representation with skew generators t_a, use instead the supplied variations $\delta A_\mu = D_\mu\varepsilon$, $\delta u = -\varepsilon^a t_a u$. Assume these variations are a gauge symmetry for every compact parameter. If the compact first variation is $\int [\langle E^\mu_A, \delta A_\mu\rangle + E_u \cdot \delta u]$, derive its covariant component identity. Here A is a connection as in Lecture 19, and the Lie algebra basis is orthonormal in its invariant pairing.

Hint. The two derivatives of the gauge parameter cancel in the covariant matter variation. In the first variation, integrate only the derivative acting on the arbitrary parameter. A radial potential has gradient orthogonal to Tu.

Answer.

$$-\partial_\mu E_A^\mu + qE_u \cdot Tu = 0.$$

In the polynomial example both $\partial_\mu E_A^\mu$ and $qE_u \cdot Tu$ equal $-2qt$. For the connection convention of part (c), the identity is $(D_\mu E_A^\mu)^a + E_u \cdot t_a u = 0$.

Exercise 19.3 (Noncommuting constant connections). Let $T_a = -\mathrm{i}\sigma_a/2$, so $[T_a, T_b] = \epsilon_{abc}T_c$. In pure Yang–Mills theory with coupling $g > 0$, take first $A_1 = T_1$, $A_2 = T_2$, $A_0 = A_3 = 0$, all constant. Compute every nonzero curvature and Euler component $E^\nu = g^{-2}D_\mu F^{\mu\nu}$, and check the covariant identity. Repeat after changing A_3 to $T_1 + T_2$. Exhibit the separate nonzero commutator contributions to $D_\nu E^\nu$ in that second case. What can these constant examples establish about an ordinary-divergence identity for arbitrary nonconstant connections?

Hint. Two raised spatial indices give no metric sign. For constant connections $E^j = g^{-2}\sum_i [A_i, [A_i, A_j]]$; compute each Lie bracket before summing.

Answer. Initially $F_{12} = T_3$ and $(E^1, E^2) = g^{-2}(-T_1, -T_2)$. After the change,

$$(E^1, E^2, E^3) = g^{-2}(-2T_1 + T_2,\ T_1 - 2T_2,\ -T_1 - T_2),$$

with divergence commutators $g^{-2}(T_3, -T_3, 0)$.

Exercise 19.4 (A local superpotential theorem and a Maxwell verification). For this exercise, assume the following local decomposition statement. It is an additional hypothesis here, not a theorem to infer from the fundamental lemma alone. For a finite-order local Lagrangian of unconstrained $\mathbb{R}^N$-valued fields on a star-shaped coordinate patch of spacetime dimension at least two, a finite-order infinitesimal gauge symmetry has a local Noether current of the form

$$j^{\mu}_{\varepsilon} = W^{\mu}_{\varepsilon} + \partial_{\nu} U^{\nu\mu}_{\varepsilon}, \qquad U^{\nu\mu}_{\varepsilon} = -U^{\mu\nu}_{\varepsilon}.$$

The current and these expressions depend linearly on the parameter and finitely many of its derivatives, with smooth finite-order local field coefficients. The remainder W vanishes when the Euler expressions and their derivatives vanish. This is a local statement on the full jet space; no global topological superpotential theorem is supplied.

Verify it directly for

$$L = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_{\mu} j^{\mu}_{\mathrm{ext}}, \qquad \partial_{\mu} j^{\mu}_{\mathrm{ext}} = 0, \qquad \delta A_{\mu} = \partial_{\mu} \varepsilon,$$

where the external current is fixed and smooth. Derive the gauge Noether current, its Euler remainder and an explicit U. On solutions, compare its spatial integral for compact spatial parameters with a parameter equal to a constant ε_{∞} outside a ball. For the latter comparison assume a smooth Maxwell field whose external current has compact spatial support on the slice and total charge $Q = \int j^{0}_{\mathrm{ext}} \mathrm{d}^3 x$. Use $F_{i0} = E_i$ and an outward sphere orientation to find the surface sign. No Schwartz or L^2 condition on the potential is imposed in this boundary comparison.

Hint. Here the density changes by the divergence of $\varepsilon j^{\mu}_{\mathrm{ext}}$. Keep this term in the first Noether current before integrating the parameter derivative off the curvature.

Answer.

$$j^{\mu}_{\varepsilon} = -F^{\mu\nu} \partial_{\nu} \varepsilon - \varepsilon j^{\mu}_{\mathrm{ext}} = -\varepsilon E^{\mu}_{A} + \partial_{\nu} (\varepsilon F^{\nu\mu}).$$

The on-shell integral is zero for compact spatial parameters and $-\varepsilon_{\infty} Q$ for the stated constant boundary value.